

1. Charge

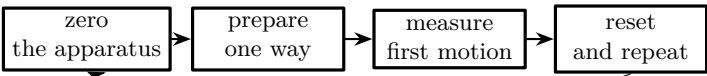
Two kinds of it, and it never appears from nowhere

Lesson 1 · two sessions of about 75 minutes · everyone builds; no electrical hazard

What you need
• Sticky tape (regular, not invisible) • Two balloons and string • Glass jar with plastic or cork lid • Stiff copper wire, about 20 cm • Aluminium foil; ruler; protractor • Plastic comb or PVC offcut • Wool sweater or silk scarf • Empty aluminium can
Predict first
Commit now. Two tape strips pulled from the same bench will ☐ attract ☐ repel ☐ do nothing. A charged balloon will pull a neutral can because _____

The rules of this test

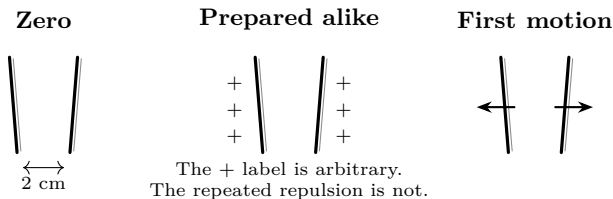
Static charge is feeble, leaky and easy to fake with a draught. That is useful: it forces good experimental habits.



1. Shut the window. Move loose paper away. Record whether the day is dry, average or humid.
2. For every tape test, use strips 15 cm long with a 2 cm folded handle. Touch the handle only.
3. Record the *first* movement. If the tapes touch, discharge both on the bench and start that trial again.
4. Repeat every claimed effect three times. One theatrical success proves nothing.

A. Find the two signs

1. Hang two unprepared strips from pencils balanced across cups. Set their lower ends 2 cm apart. Wait ten seconds. This is the zero trial.
2. Press two fresh strips separately onto the same clean bench. Smooth each ten times. Pull each off briskly and hang it at once.
3. Begin 8 cm apart. Bring the stands inward 1 cm every three seconds. Write the gap at the first repeatable motion and its direction.
4. Discharge, repeat twice, and swap the stands left-to-right for trial 3. Did the result follow the tape or the side of the room?

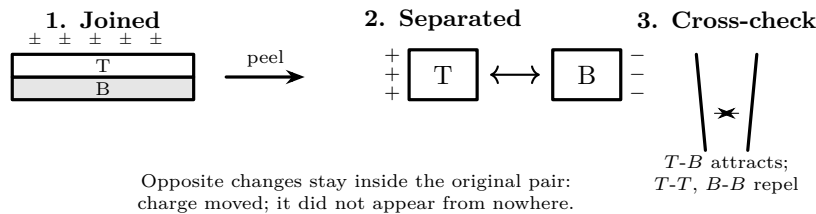


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B. Give two tapes opposite histories

1. Put a fresh strip on the bench and mark its handle B , for bottom. Press a second strip on top and mark it T , for top.
2. Lift the joined pair by B . Before peeling, answer: is the pair as a whole charged, neutral, or not yet known?
3. Peel T from B in one clean pull. Hang both at once. Record their first motion from 8, 7, 6, ... cm.
4. Make a second pair. Compare T_1 with T_2 , then B_1 with B_2 , resetting between comparisons.



Pair	Trial	First motion	Threshold gap	Anything touched?
same-bench strips	1		cm	
	2		cm	
	3, sides swapped		cm	
T_1-B_1	1		cm	
T_1-T_2	1		cm	
B_1-B_2	1		cm	

Stop and argue

Attraction cannot, by itself, prove opposite net charges: a charged object can attract a neutral one. Which tape result *cannot* be explained by a neutral partner? Why is swapping the stands a useful control?

C. Polarisation: water and aluminium

1. Lay the empty can on a level bench and mark its starting line. With the balloon uncharged and 2 cm away, verify that the can stays put.
2. Rub the balloon on wool exactly ten strokes. Approach the can from 12 cm in 1 cm steps, pausing three seconds. Never touch it.
3. Record the first gap that starts the can rolling. Recharge with ten strokes and repeat twice. Then touch the balloon to the bench and repeat as a discharged control.
4. Run a water stream about 2 mm wide. Measure the sideways deflection 5 cm below the tap with the charged balloon held 3 cm away.

5. Discharge the balloon and repeat. Do not report “it bent”; report direction and millimetres.



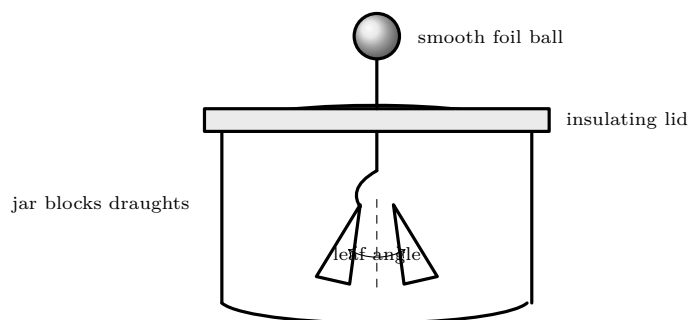
Trial	Charged can gap	Discharged gap	Water deflection
1	cm	cm / none	mm toward / away
2	cm	cm / none	mm toward / away
3	cm	cm / none	mm toward / away

The question that matters

The can began neutral. If it still moved toward the balloon, attraction is not proof of opposite net charge. The nearer charges rearranged or turned; the farther ones could not cancel the stronger near-side interaction.

D. Build an electroscope

1. Ask the adult to make a close-fitting hole through the lid. Push 20 cm of stiff copper wire through it. The wire must not touch a metal lid.
2. Bend the inner 3 cm into a narrow hook. Cut two identical foil leaves, each 1×4 cm. Round their lower corners.
3. Hang both leaves on the hook so their flat faces touch. Crumple foil tightly around the outer end to make a smooth 2 cm ball.
4. Close the jar. The leaves must hang freely, clear of glass and lid. Tape a paper protractor behind the jar with 0° on the centreline.
5. Zero test: touch the ball for three seconds. Wait ten seconds. Record the resting separation angle. It should be 0° or close to it.

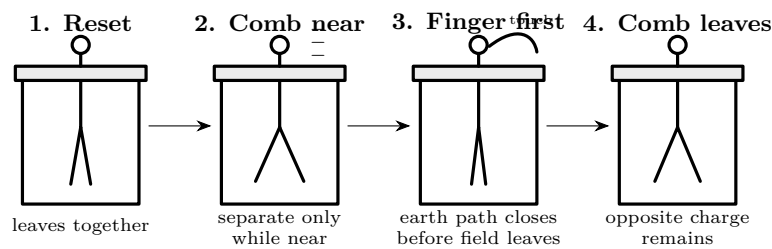


BUILD CHECK

The two leaves are identical, the wire is bare where the foil touches it, the wire touches no metal lid, and the leaves touch nothing except the hook.

E. Make three distinct states

- Contact charge.** Rub the comb ten strokes. Touch it to the ball for one second; remove it. Record leaf angle at 0, 10, 30 and 60 seconds.
- Touch the ball with one finger for three seconds. The leaves should collapse. This is the reset, not an optional flourish.
- Near, no touch.** Recharge the comb. Hold it 2 cm from the ball. Record the leaf angle. Withdraw it. Record the angle again.
- Charge by induction.** Bring the comb back to 2 cm. While it stays there, touch the ball. Remove your finger *first*; remove the comb second. Record the final angle.



State	0 s	10 s	30 s	60 s	Returns after removal?
contact charge	◦	◦	◦	◦	
comb near, no touch	◦				yes / no
finger then comb removed	◦	◦	◦	◦	yes / no

The proof, not the performance

- Two signs.** Like-history pairs repelled in ____/3 trials. The opposite-history matrix showed _____. Repulsion rules out a neutral second object.
- Conservation.** *T* and *B* began as one system and acquired opposite behaviours together. State what moved and where the balancing charge remained: _____
- Polarisation.** The neutral can began moving at a median gap of ____ cm; the discharged control _____. Therefore attraction alone does not identify net sign.
- Conduction.** Contact left a leaf angle after the comb departed; the near-only state did not. The difference is charge transfer.
- Induction.** Removing the earth path first trapped a final leaf angle of ____°. Reversing the removal order should erase it. Do that control once.

Evidence chain
Repulsion identifies like signs. The <i>T/B</i> matrix identifies opposite histories. The discharged balloon is the null control. Persistent leaf angle distinguishes transferred charge from temporary rearrangement. Removal order tests the induction explanation.

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When it refuses to work

Symptom	Find the cause	Change one thing
Leaves never move	Damp foil, dirty insulator, weak charge	Dry jar and lid; shorten preparation-to-test delay.
Leaves stick to glass	Hook off-centre or leaves too long	Re-centre wire; trim identical leaves.
Leaves move before comb arrives	Draught or charged jar wall	Discharge all surfaces; close jar; wait.
Angle falls quickly	Humidity or leakage across lid	Record decay; clean and dry lid; repeat.
Induction leaves no charge	Wrong removal order	Say aloud: finger first, comb second.

Five questions before you claim victory

Which observation proves like signs? Which control separates charge from a draught? Why does the neutral can move? Why must finger-before-comb matter? Which measured result was least repeatable, and what limits it?

What it means

Charge has two signs. Like signs repel; unlike signs attract. Rubbing and peeling transfer electrons between objects, leaving equal and opposite changes inside the larger system. Conductors let charge travel; insulators slow that travel. A nearby charge can rearrange charge without touching. The electroscope makes each of those sentences visible, but only the controls make them evidence.

The link to the coil

The topload of a Tesla coil is a smooth conductor that stores charge. A streamer is charge escaping into air. Your electroscope shows the same family of effects at a vastly smaller energy. Similar physics is not similar risk.

This lesson is low voltage. Everything here runs from batteries or a small plug-in supply and cannot hurt you. That changes at Lesson 10. Build the habits now — one hand, discharge before touching, check before you power up — while the stakes are nothing, so that they are automatic when the stakes are real.

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